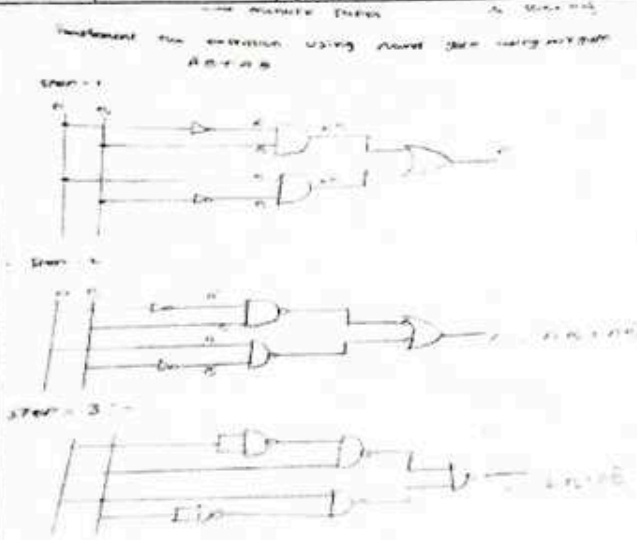




Degree, Semester & Branch : III Semester B.E. ECE A
Course Code & Title : EC3352 & DIGITAL SYSTEM DESIGN
Name of the Faculty member: Mrs.G.Subhashini

Sl.No.	Date	Topic(s)	Activity*	Reference
UNIT I BASIC CONCEPTS				
1.	25.09.2023	Implementation of Boolean expressions using universal gates	One Minute Paper	https://oncourseworkshop.com/self-awareness/one-minute-paper/
				
		One minute Paper P. Kishore Implement the expression using • NAND gate by using $\overline{AB} + AB$ Step 1: Draw AOI logic Step 2: Draw bubbles on o/p of AND gate and I/p of OR gate Step 3: Replace by NAND-gate		
				


Signature of the Faculty Member


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Department of Electronics and Communication Engineering

Academic Year: 2023 – 2024(ODD Semester)

INNOVATIVE TEACHING METHODS

Degree, Semester & Branch : III Semester B.E. ECE A

Course Code & Title : EC3352 & DIGITAL SYSTEM DESIGN

Name of the Faculty member: Mrs.G.Subhashini

Sl.No.	Date	Topic(s)	Activity*	Reference
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UNIT II COMBINATIONAL LOGIC CIRCUITS

1.	19.10.2023	Parity Generator/Checker, Seven Segment display decoder	Note Check	https://www.niu.edu/citl/resources/guides/active-learning-activities.shtml#:~:text=Note%20Check,clarification%20on%20a%20course%20concept.
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Truth Table

X	Y	Z	Odd parity	Even parity
0	0	0	1	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	0	1

K-map

Odd parity

$$\text{odd} = \bar{x}y\bar{z} + x\bar{y}\bar{z} + x\bar{y}z + x\bar{y}z + \bar{x}(y\bar{z} + yz) + \bar{x}(y\bar{z} + yz) + x(y\bar{z} + yz) + x(y\bar{z} + yz)$$

Even parity

$$\text{even} = \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}y\bar{z} + \bar{x}y\bar{z} + \bar{x}(y\bar{z} + yz) + \bar{x}(y\bar{z} + yz) + x(y\bar{z} + yz) + x(y\bar{z} + yz)$$

P	A	B	C	Parity Even PBC
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	1	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

G. Subhashini
9/10/23

Signature of the Faculty Member

HOD
10/11/23
HOD



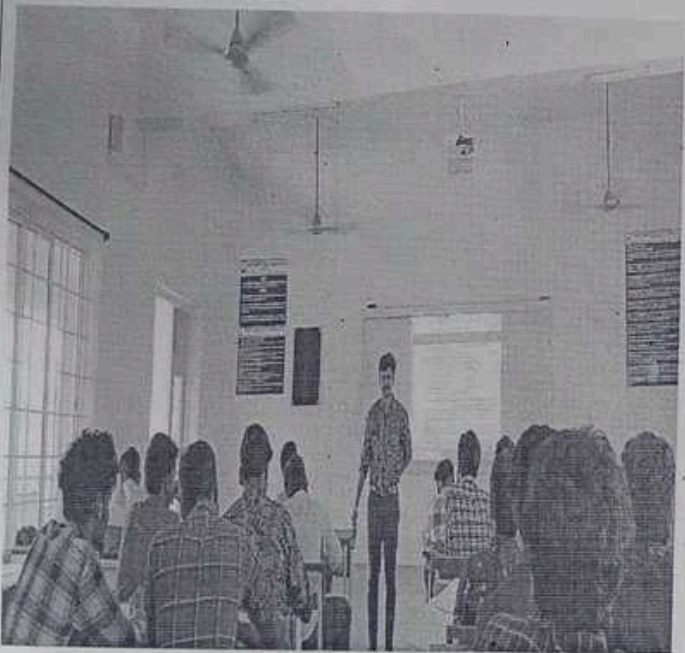
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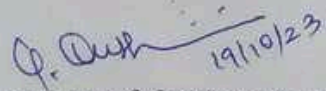
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Degree, Semester & Branch : III Semester B.E. ECE A
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Name of the Faculty member: Mrs.G.Subhashini

Sl.No.	Date	Topic(s)	Activity*	Reference
UNIT III SYNCHRONOUS SEQUENTIAL CIRCUITS				
1.	11.11.2023	Design – Moore & Mealy models	Zero-Minute Speech	https://cft.vanderbilt.edu/wp-content/uploads/sites/59/Active-Learning.pdf
				


Signature of the Faculty Member


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RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2023 - 2024 (ODD Semester)

Degree, Semester & Branch: III Semester B.E. ECE A

Course Code & Title: EC3352 & Digital System Design

Name of the Faculty member: Mrs.G.Subhashini, AP/ECE

Topic: Hazards

ACTIVITY: Flipped Classroom
Lesson Plan

Key elements of Flipped Classroom	• To provide an opportunity for students to gain first exposure to content prior to class. • To provide an incentive for students to prepare for class. • To provide in-class activities that focus on higher-level cognitive activities. • To provide a mechanism to assess student understanding.
Topic	Hazards, Essential Hazards
Justification for Choosing the Topic using Flipped Class	The topic was chosen for flipped class as there are many interesting factors about Hazards. The Unit 4 deals with the details of Asynchronous Sequential Circuits and functionalities, various techniques used for hazards and Hazards free circuits.. Since the students are exposed to the details of Hazards, I taught of giving the topic 'Hazards and Essential Hazards' for flipped class. This topic can be learnt in self-learning mode. ❖ As there are many problems can be solved in Hazards, few problems were discussed in their prescribed text book. If this flipped class is used for the topic apart from the learning materials given by the faculty, students have chance to browse and find more information related to specific application. This is the reason for choosing this topic for flipped class activity.

Subject	Digital System Design
Time Needed	At Home: 10 minutes to watch each video, total of 20 minutes Students will be assigned videos two days in advance to give them time to watch them at home and complete At-Home Activities In class: 50 minutes class period
Learning Objective	To explain the concepts of Hazards in various types like Static Hazard, Dynamic Hazard and Essential Hazard
Expected Outcome	At the end of the flipped class activity, ❖ Students will be exposed to collaborative learning environment. ❖ The students will develop better understanding of the concepts and principles. ❖ The students will be able to explain the various applications of Hazards with realtime examples in the areas of Digital Signal Processing, VLSI, Communication and Embedded system Circuits. ❖ The problem solving skills of the students will be improved. ❖ It will help students to perform better at end semester examination.
Student Learning Resources(Home) ie: videos, website, teachercreated materials	https://archive.nptel.ac.in/courses/108/105/108105132/ The NPTEL video discusses about Hazards in detail and shows the problems related to Hazard free circuit. The NPTEL video gives the detailed presentation of Hazards. Kindly make use of book chapters for preparing the presentations and apart from the details any other information regarding the applications can also be added from other online resources with proper citations. Prepare a presentation for 7 minutes containing the ✓ Hazards ✓ Types of Hazards ✓ Static Hazard ✓ Dynamic Hazard ✓ Essential Hazard ✓ Hazard free circuits

	❖ The book chapters will be shared to the students as materials and video like from NPTEL will be posted related concept of Hazards in the course website. ❖ The learning material of book chapters consist of details of Hazards for applications such as ❖ Hazard assessments are simply a process of identifying hazards, ❖ Evaluating the risks presented by those hazards, ❖ Managing the risks of the hazards of the experiment to be performed by incorporating appropriate hazard controls into the experimental design process.
Student Learning Activities(Home)	1. Students will watch 2 YouTube videos about the Hazards . 2. While watching the videos, students will note the important information on the specified topic to help them collect information and write questions that they still have. 3. Reading of learning material: 10 days time duration 4. Preparation of presentation: 4 days time duration 5. Discussions and clarification of doubts: Any time within the 2 weeks time for discussion within the group mates and any clarification regarding the flipped class
Classroom Activities and Materials	1. Presenting the topic assigned by each group – 5 minutes duration 2. Question & Answering and Oral feedback: Class will open with a question and answer session and the teacher will ask questions raised by students if no one provides questions in class. These will serve as the discussion.– 2 minutes duration

Implementation process:

- ❖ The students were informed about the flipped class activity atleast 2 weeks before the implementation of the activity.
- ❖ The materials like documents, videos related to the topic will be posted in the course website - Canvas.
- ❖ The grouping of students will also be done well in advance and each group is allotted with one topic of application of their choice among the 6 applications listed. (2 weeks before implementation)
- ❖ An announcement will be posted once in two days about the activity as reminder.
- ❖ A Discussion forum will be opened for flipped class, in which the group members can

- use for discussion outside class. Using the 'your questions about flipped class' – doubts of the students can be posted and the instructor will clarify within the 8 hours of post.
- ❖ A whatsapp group will be created for flipped class activity to clarify the doubts and discussions.
 - ❖ The students have to prepare a presentation for 5 minutes with the concept learnt and 2 minutes for Question & answering and tell about the contribution of each member.
 - ❖ The inside class activity: presentation will be planned for 1 period. In which each group has to present their topic chosen and others will also listen to the presentation.
 - ❖ Time management: The flipped class activity should be able to complete in the planned duration.
 - ❖ The concepts of Hazards are covered to the whole class and the handouts prepared on the topic will be shared within the group members.
 - ❖ Each group will be assessed through the content of presentation, communication skills, Q& A, contribution by each member as shown in the table below. Both formative and summative assessment will be made.
 - ✓ Content of presentation (20 Marks)
 - ✓ Communication skills (10 Marks)
 - ✓ Q & A (10 Marks)
 - ✓ Contribution (10 Marks)
 - The main objective of the flipped class activity is each group will learn one application and presentation helps the other members also to know about all the application topics.
 - Thus, some of the concepts of Hazards will be covered in 1 period through this activity.
 - After completion of presentation, both oral and written feedbacks will be collected about the flipped class from the students.
 - The evaluation on the flipped class is done finally and will be assessed based on the following parameters. The marks will be considered for final internal mark calculation.

Student Feedback to access the success of the flipped class:

The feedback about the success of the flipped class will be collected in two modes

- ❖ **Oral feedback**
- ❖ **Written Feedback**

Oral Feedback:

The oral feedback will be collected at the end of the presentation by the students about

- ✓ Experience in doing this flipped class
- ✓ Contribution by each member of the group.
- ✓ What was the learning outcome of such activity?
- ✓ What is the difference you felt between traditional teaching and flipped class?

Written Feedback:

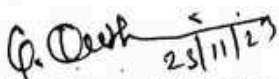
A written feedback will be collected with the following questionnaire at the end of completing the activity.

Students Feedback: Flipped Class Room Activity

1. The Flipped classroom is more engaging than traditional classroom instruction.?
☐ Disagree ☐ Agree ☐ Neither agree or disagree
2. I like watching the lessons on video.?
☐ Disagree ☐ Agree ☐ Neither agree or disagree
3. Learning how to use a Flipped Classroom will benefit me in my future education.
☐ Disagree ☐ Agree ☐ Neither agree or disagree
4. The Flipped Classroom gives me greater opportunities to communicate with other students.
☐ Disagree ☐ Agree ☐ Neither agree or disagree

Name & Signature of the Student with date

Based on the feedback collected, the success of the activity can be assessed and from the challenges faced during preparation the problems can be identified and rectified in the future flipped class activities.


Signature of the Faculty


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Academic Year: 2023 - 2024 (ODD Semester)

Degree, Semester & Branch: III Semester B.E. ECE A

Course Code & Title: EC3352 & Digital System Design

Name of the Faculty member: Mrs.G.Subhashini

Topic: Comparison of Logic families

Learning Outcomes:

The students will be able to

- ❖ Represent the Logic families using basic circuits
- ❖ Differentiate Resistor Transistor Logic, Transistor Transistor Logic, Emitter Coupled Logic and CMOS
- ❖ Participate actively in the learning process.
- ❖ Discuss the concepts, convey ideas and share the views to really understand the different Logic families.
- ❖ Communicate effectively by sharing their views among the teammates.

Justification for choosing the topic for the activity:

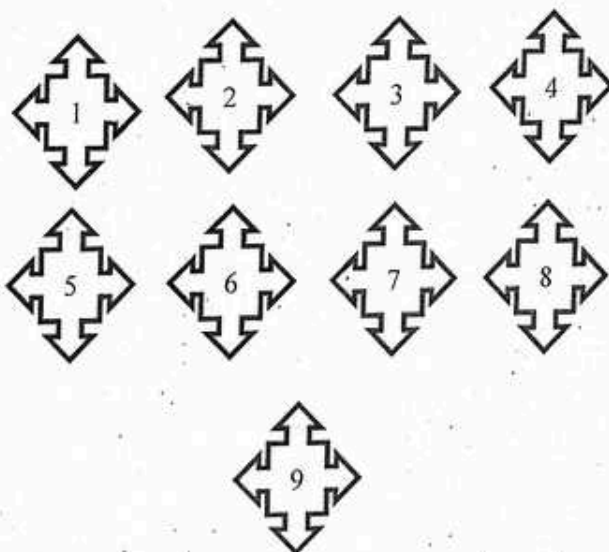
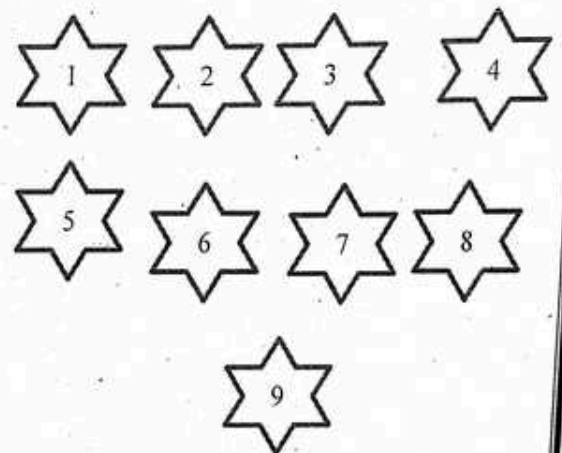
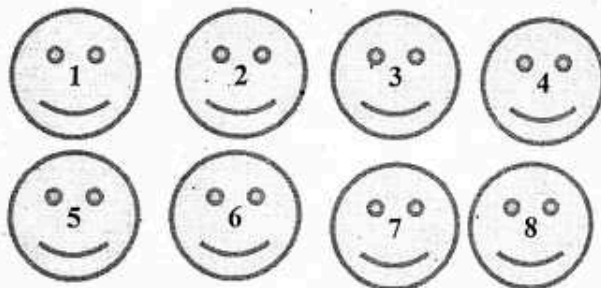
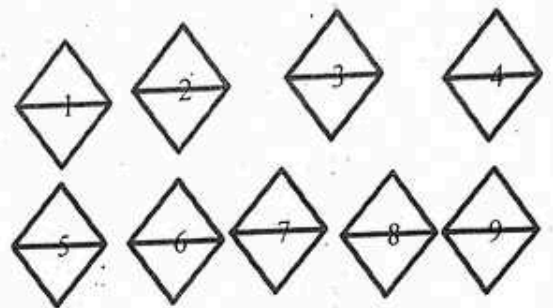
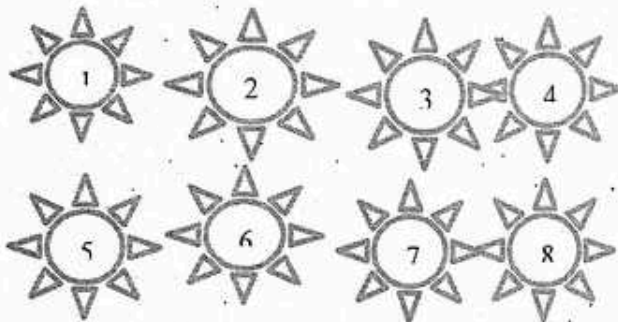
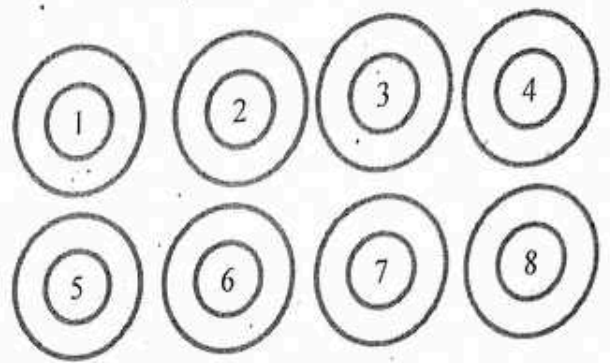
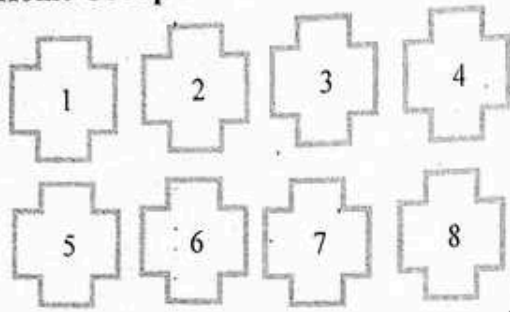
The concept of are already learnt in Electronic Devices and circuits. This topic is related to the same concepts and easy to understand. The activity '**JIGSAW**' is used to recall the concepts learnt and enhance the learning through this collaboration.

Plan for Implementation:

- ❖ The material related to the topic, video will be posted 1 week before the conduct of activity.
- ❖ **Activity: JIGSAW**- The Class Strength is divided into 7 Shapes Groups with 8 members and 3 groups consists of 9 members as shown below using Random Lot System.
- ❖ Each group is allotted with a shape and its known as **Home group** as shown below.
- ❖ 7 Shapes are named 1,2,3,4,5,6,7,8. All the 1's in different Shapes are made to form Expert group.
- ❖ Each Expert group is allotted with one topic
 - ✓ Introduction of Resistor Transistor Logic
 - ✓ Introduction of Transistor Transistor Logic
 - ✓ Introduction of Emitter Coupled Logic & Introduction of CMOS
 - ✓ Working of Resistor Transistor Logic
 - ✓ Working of Transistor Transistor Logic
 - ✓ Working of Emitter Coupled Logic
 - ✓ Working of CMOS
 - ✓ Comparison of TTL,RTL,ECL and CMOS
- ❖ The expert group members discuss the topic in detail for 20 minutes with the learning materials already posted in the course website – canvas.
- ❖ The Expert group members should go to their original shape group and discuss the points (30 Minutes) to the other members.
- ❖ All the members in home group learnt about all the topics through expert group members.
- ❖ Finally one from each shape group should summarize the points of the topic Triangular and saw tooth wave generators to the class (15 Minutes).
- ❖ Review of points and conclusion of the activity (5 minutes)

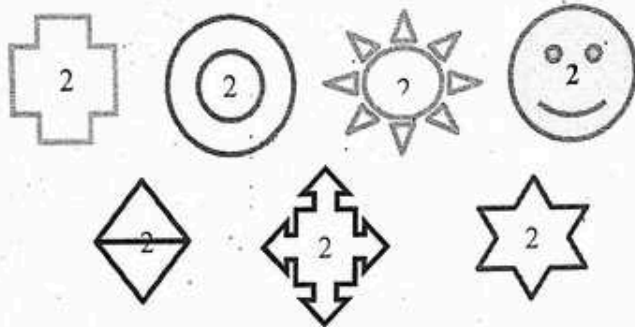
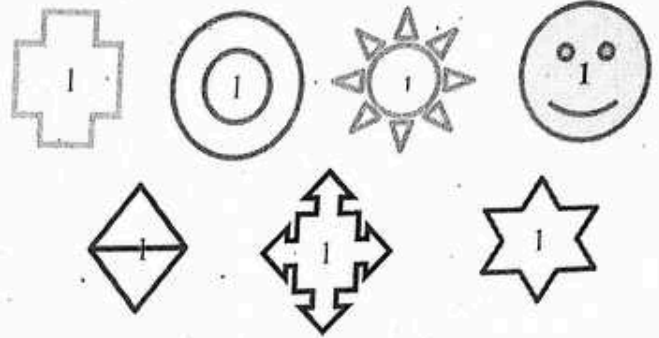
❖ Thus the concept of Logic families is known to all by involving the students actively.

Home Group formation:



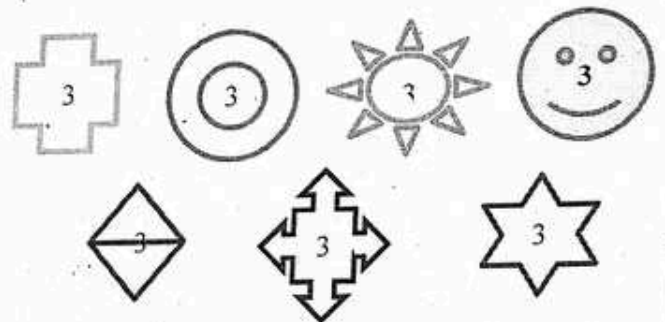
GROUP 1

✓ *Topic:* Introduction of Resistor
Transistor Logic



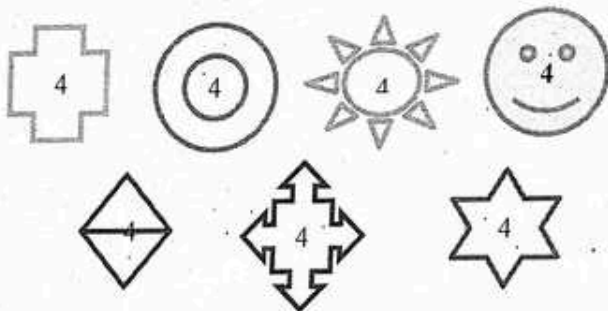
GROUP 2

Topic: Introduction of
Transistor Transistor Logic



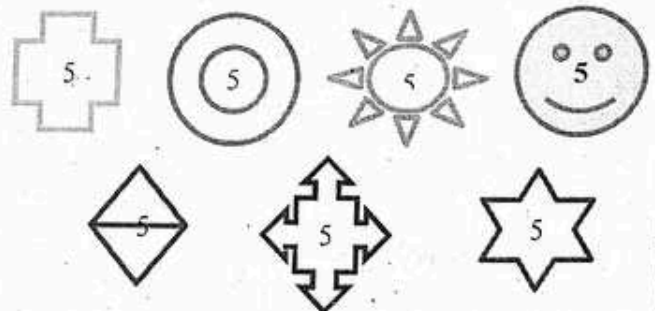
GROUP 3

Topic: Introduction of Emitter
Coupled Logic & Introduction
of CMOS



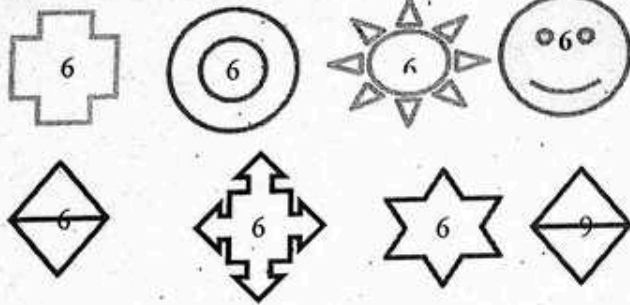
GROUP 4

✓ *Topic:* Working of Resistor
Transistor Logic



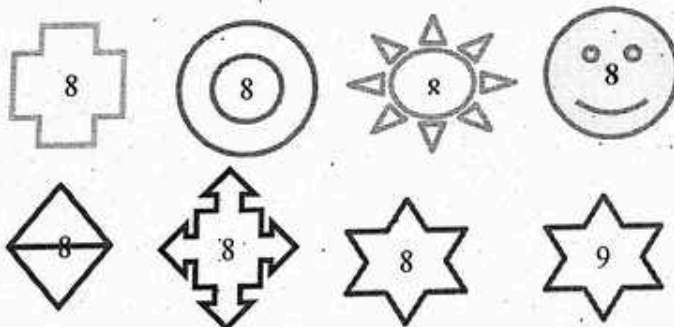
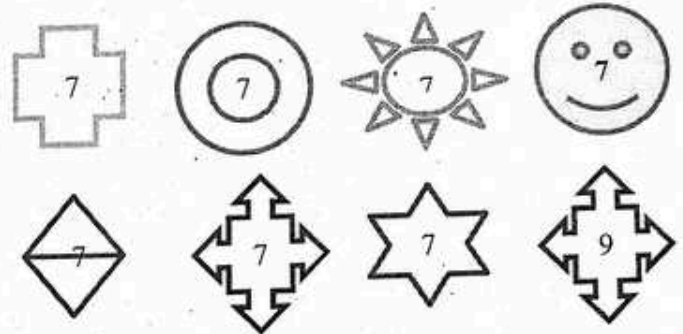
GROUP 5

Topic: Working of Transistor
Transistor Logic



GROUP 6
✓ Topic: Working of Emitter
Coupled Logic

GROUP 7
✓ Working of CMOS



GROUP 8
✓ Comparison of TTL,RTL,ECL
and CMOS (Advanced
Learners)

References:

Kindly refer the following materials before coming to the Class

- <https://www.jigsaw.org/>
- https://www.newcastle.edu.au/data/assets/pdf_file/0016/109600/Jigsaw-learning-activity.pdf

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12/12/23
Signature of the Faculty

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